

2G wireless technologies in most radio conditions, but is misinterpreted by TCP as network congestion.

Finally, Wireless Session Protocol (WSP) is best thought of on first approach as a compressed version of HTTP. This protocol suite allows a terminal to emit requests that have an HTTP or HTTPS equivalent to a WAP “gateway”; the gateway translates requests into plain HTTP.

In the Wireless Application Environment (WAE), application-specific mark-up languages are defined. The primary language of the WAE is WML, the Wireless Mark-up Language, which has been designed from scratch for handheld devices with phone-specific features. WML is an XML-compliant format. However, since XML documents can take up a lot of room, a specific compression technique for XML documents was developed (wireless binary XML, or WBXML).

There has been considerable discussion about whether the WAP protocol design was appropriate. The initial design of WAP was specifically aimed at protocol independence across a range of different protocols (SMS, IP over PPP over a circuit switched bearer, IP over GPRS etc). This has led to a protocol considerably more complex than an approach directly over IP might have caused.

Most controversial, especially for many from the IP side was the design of WAP over IP. WAP’s transmission layer protocol, WTP, uses its own retransmission mechanisms over UDP to attempt to solve the problem of TCP’s inadequacy for high packet loss networks.

The new version of WAP, WAP 2.0, is a re-engineering of WAP using a cut-down version of XHTML with end-to-end HTTP, i.e. dropping the gateway and custom protocol suite used to communicate with it. Some observers predict that this next-generation WAP will converge with, and be replaced by, true Web access to pocket devices. Whether this next generation